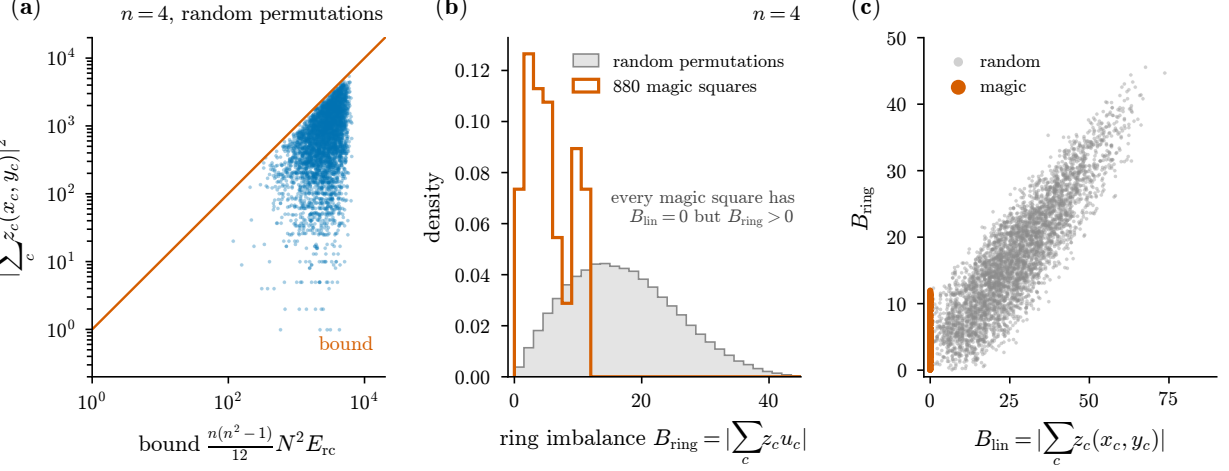

(a)
$n = 4$, random permutations
$|\sum_c z_c(x_c, y_c)|^2$
bound
bound $\frac{n(n^2-1)}{12} N^2 E_{\mathrm{rc}}$
(b)
$n = 4$
random permutations
880 magic squares
every magic square has
$B_{\mathrm{lin}} = 0$ but $B_{\mathrm{ring}} > 0$
density
ring imbalance $B_{\mathrm{ring}} = |\sum_c z_c u_c|$
(c)
random
magic
B_{ring}
$B_{\mathrm{lin}} = |\sum_c z_c(x_c, y_c)|$